

# DAVIS: A Depth-Only End-to-End Active-Vision Framework for Humanoid Soccer Skills

Anonymous Author(s)

Affiliation

Address

email

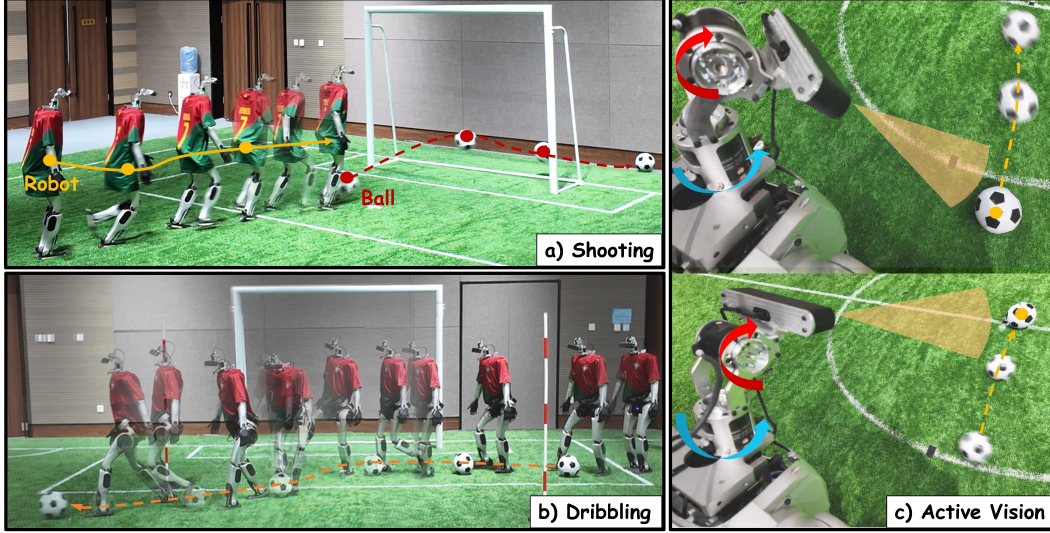


Figure 1: **Overview of our humanoid soccer system.** (a) The robot performs goal-oriented shooting by approaching the ball and kicking it into the goal. (b) The robot demonstrates agile multi-directional dribbling with continuous ball control, including turning and weaving around poles. (c) Our active vision design enables the robot to actively reorient its head in yaw and pitch to keep the ball near the center of the field of view, thereby improving perception-action coordination.

**Abstract:** Humanoid soccer contact skills require more than producing high-impact foot-ball contacts: the robot must close the loop over perception, approach, alignment, impact, and recovery while its own motion induces substantial view-point changes, frequent loss of the ball from view, and uncertain contact outcomes. In this work, we ask a compact yet stricter question: can a humanoid learn soccer contact skills using only a head-mounted depth image, proprioceptive history, and an optional low-dimensional task command, and directly output 25-DoF joint PD targets without extra runtime perception or planning modules? To this end, we propose **DAVIS**, a depth-only end-to-end framework for humanoid soccer skills that learns visibility-aware auxiliary geometry during training, and combines GT-to-prediction annealing, task curricula, and AMP-style motion priors to smoothly bridge privileged supervision and real deployment. Built on this framework, we instantiate representative soccer contact skills, including goal-directed shooting and directional dribbling, through task-specific definitions of objects, commands, rewards, and curricula, and validate them through simulation, Noetix E1 real-robot experiments, and ablations.

**Keywords:** Humanoid Soccer, Active Vision, Reinforcement Learning

# 1 Introduction

Humanoid robots have made substantial progress in reinforcement-learning-driven dynamic locomotion [1, 2, 3, 4, 5, 6], but soccer remains a harder setting because it requires the robot to perceive, approach, align, contact, and recover while the visual target and the robot itself are both moving. Unlike standard locomotion benchmarks, soccer contact skills require the policy to reason not only about *where the ball is*, but also about *how to approach it*, *how to keep the target in view*, and *how contact should change the subsequent ball motion*.

On real humanoids, onboard sensing and whole-body control further amplify these difficulties: the head-mounted view changes with body motion, active head yaw and pitch determine what evidence the policy receives, and small perception or stance errors can change the contact outcome [7, 8, 9, 10, 11]. Existing soccer systems often manage this complexity with modular perception, localization, planning, and skill-switching interfaces, including RGB detectors, LiDAR/RGB-D-based localization, or external state inputs. While practical, these interfaces introduce runtime dependencies whose errors can propagate when visibility, posture, and contact timing change together. As humans can infer task geometry from only egocentric vision, we raise the question: **can a humanoid learn deployable soccer contact skills from only head-mounted depth, proprioceptive history, and low-dimensional task commands — without any external scene state?**

To answer this question, we propose **DAVIS**, a depth-only end-to-end structured reinforcement learning framework. At deployment, the policy remains a single closed-loop mapping from head-mounted depth, proprioceptive history, and low-dimensional commands to 25-DoF joint PD targets. This is achieved by placing structure inside the learning signal rather than in the deployment pipeline, following the broader idea of using privileged simulation information to guide deployable policies [12, 13, 14]. Specifically, auxiliary geometry and visibility predictions are learned from privileged simulation supervision, while the actor gradually transitions from stable geometric guidance to self-predicted features via GT-to-prediction annealing and visibility gating. With this interface fixed, different soccer behaviors can be learned simply by changing the local objective, geometry, and supervision that define the task.

The many skills that can be defined in this setting far exceed the two considered here. With the deployment interface fixed, different soccer skills are distinguished by their task definitions rather than by additional runtime modules. We focus on goal-directed shooting and directional dribbling because they represent two different contact regimes: one-shot goal-directed striking and sustained direction-conditioned ball control. In both cases, task variables, rewards, and curricula define the corresponding task target, while the depth-only pipeline stays unchanged; other behaviors such as passing, obstacle-guided dribbling, or ball re-search after loss of sight can be obtained in the same way by redefining the local geometry and supervision.

In summary, our contributions are as follows: (1) We introduce a **depth-only end-to-end RL framework** for humanoid soccer contact skills, where the deployed policy maps head-mounted depth, proprioceptive history, and low-dimensional task commands directly to 25-DoF joint targets with active-vision head control. (2) We develop a **visibility-aware auxiliary geometry mechanism** that learns task-relevant object geometry from the depth latent, and combines visibility masking with GT-to-prediction annealing to stabilize visual learning without adding runtime perception modules. (3) We instantiate the framework on **penalty kick and directional dribbling** to validate its feasibility and generalization across representative soccer contact skills, and close the loop from IsaacLab training to E1 real-robot deployment with depth alignment, policy export, and 25-DoF PD control.

## 2 Related Work

**Robot Soccer Skill Learning.** Robot soccer has long served as a challenging benchmark for robot control, requiring the integration of visual perception, real-time decision making, agile locomotion, and ball-oriented skills [15, 16, 17]. RoboCup-style soccer systems often relied on modular perception-control pipelines, where sensing, state estimation, and soccer control were separately engineered rather than learned end-to-end [18]. Recent learning-based methods have enabled legged

robots to acquire soccer skills through various paradigms, including reinforcement learning and imitation learning [19, 20, 21, 22, 23, 24, 25, 26]. Yet most existing systems still depend on RGB-based object detection or multi-sensor perception pipelines [27, 28, 29, 30]. In contrast, our work studies depth-driven humanoid soccer skill learning, where the robot learns to chase, dribble, and kick directly using only depth observations and proprioception.

**Whole-Body Loco-Manipulation for Humanoid Soccer.** Whole-body loco-manipulation on humanoid robots is inherently challenging, as the robot must simultaneously coordinate balance, locomotion, and task-directed manipulation within a high-dimensional and tightly coupled control space [31, 32, 33]. To make the problem tractable, prior systems often introduce decoupled or hierarchical interfaces, such as separate upper- and lower-body controllers or high-level task policies paired with low-level whole-body controllers [34, 35, 36, 37, 38]. Humanoid soccer further amplifies this difficulty: unlike manipulating static objects, the ball is a highly dynamic target, requiring the robot to approach from a suitable direction, maintain visibility, and generate precise foot-ball contacts for effective ball control. To address this challenge, we propose a whole-body control framework, **DAVIS**, that jointly optimizes active visual tracking, approach, and foot-ball interaction for dynamic ball-centric loco-manipulation.

## 3 Method

### 3.1 Problem Formulation

We model humanoid soccer contact skills as a POMDP problem [39], denoting the process as  $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma \rangle$ . The simulator maintains a full state  $\mathbf{s}_t \in \mathcal{S}$  containing robot, ball, goal, and contact-related physical variables; during training, privileged variables derived from this state, such as ball pose, goal pose, visibility, contact state, and reference motion, are used as critic inputs, reward-computation terms, and auxiliary-supervision labels. The deployed policy instead observes only a head-mounted depth image  $\mathbf{D}_t$ , a proprioceptive history  $\mathbf{p}_{t-H:t}$  of length  $H$ , and an optional low-dimensional task variable  $\mathbf{c}_t$ ; for instances without an external command,  $\mathbf{c}_t$  is omitted. We write the deployed observation and action as  $\mathbf{o}_t = (\mathbf{D}_t, \mathbf{p}_{t-H:t}, \mathbf{c}_t)$  and  $\mathbf{a}_t = \pi_\theta(\mathbf{o}_t) \in \mathbb{R}^{25}$ . The proprioceptive history is encoded inside the policy to infer latent body dynamics, while all explicit task geometry used by the actor must be produced from deployable inputs. The low-level controller converts the action into joint PD targets  $\mathbf{q}_t^{\text{des}} = \mathbf{q}^0 + \mathbf{s} \odot \mathbf{a}_t$ , where  $\mathbf{q}^0$  is the nominal joint configuration and  $\mathbf{s}$  is the per-joint action scale. The E1 humanoid consists of 23 body DoFs and 2 head DoFs, so we decompose the action as  $\mathbf{a}_t = [\mathbf{a}_t^{\text{body}}, \mathbf{a}_t^{\text{head}}]$  with  $\mathbf{a}_t^{\text{body}} \in \mathbb{R}^{23}$  and  $\mathbf{a}_t^{\text{head}} \in \mathbb{R}^2$ . Because the depth camera is mounted on the actuated head, the head action is part of the sensing loop: it directly changes the next depth observation. We optimize the asymmetric actor-critic policy with PPO [40] by maximizing the expected discounted return  $\max_\theta \mathbb{E}_{\pi_\theta} \left[ \sum_{t=0}^T \gamma^t (r_t^{\text{task}} + \lambda_{\text{amp}} r_t^{\text{amp}} - p_t^{\text{reg}}) \right]$ , where  $r_t^{\text{task}}$  is the task reward,  $r_t^{\text{amp}}$  encourages motion patterns close to a reference prior, and  $p_t^{\text{reg}}$  penalizes joint limits, action changes, unstable posture, collisions, and unsafe contacts.

### 3.2 Depth-Only End-to-End Structured Soccer RL Framework

We build **DAVIS** around a deployment interface that remains deliberately narrow: the actor receives head-mounted depth, proprioceptive history, and an optional low-dimensional command, and outputs 25-DoF joint target residuals for the PD controller, as illustrated in Fig. 2. During training, privileged simulation states provide dense supervision for perception, value learning, and motion regularization; at deployment, the policy keeps only the depth/proprioception-to-control path.

To keep this deployment path compact, we introduce task structure only through training-time supervision and regularization. The depth latent feeds auxiliary geometry and visibility heads, whose predictions are confidence-gated before entering the actor. Simulation ground truth supplies geometry and visibility labels, but only as auxiliary supervision and as early-stage geometric guidance. A GT-to-prediction annealing schedule gradually replaces privileged features with predicted features, so the actor first learns how local geometry should affect whole-body motion and then operates with geometry inferred from depth. In parallel, PPO optimizes task return, the history estimator learns

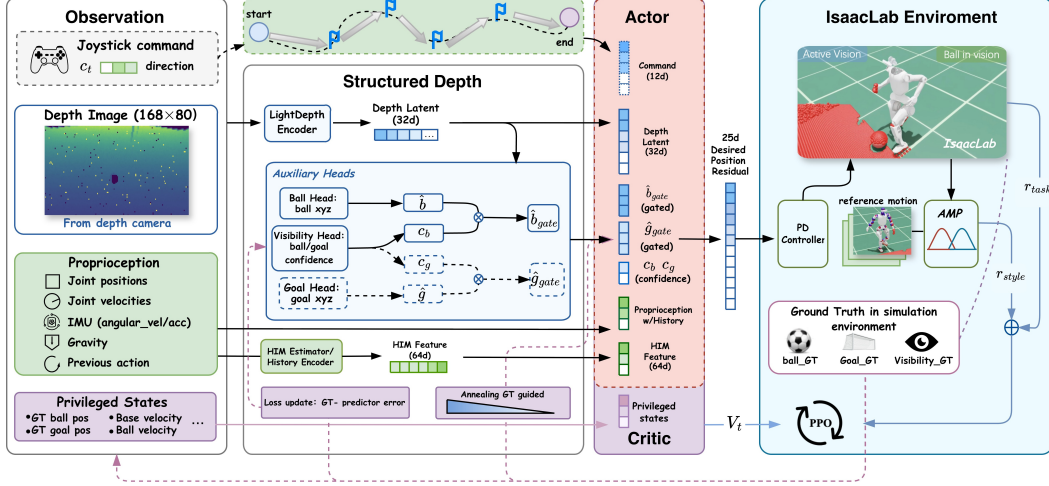


Figure 2: **Overview of our DAVIS framework.** The observation stream contains the command  $c_t$ , a single  $168 \times 80$  depth image, proprioceptive history, and privileged simulation states. The structured-depth module encodes the depth input into a 32-D latent, predicts ball/goal geometry and visibility confidence through **auxiliary heads**, forms confidence-gated geometry features, and combines them with a 64-D HIM/history feature. The actor receives the command, depth latent, gated geometry/confidence features, proprioceptive history, and HIM feature, and outputs a 25-D position residual for the PD controller. The auxiliary losses, GT-to-prediction annealing, PPO update, task reward, and style reward use simulation-only signals from IsaacLab rollouts, including ball, goal, and visibility ground truth. Dashed boxes in the structured-depth module indicate optional task-specific modules.

117 proprioceptive state inference, and AMP-style motion priors regularize full-body motion. These  
 118 components make long-horizon soccer contact learning tractable while preserving a single deploy-  
 119 able policy graph [41, 42, 43].

120 This framework separates reusable learning machinery from task definition. The reusable part con-  
 121 sists of the depth/proprioceptive encoders, auxiliary geometry and visibility learning, Asymmetric  
 122 actor-critic optimization, motion-prior regularization, and depth alignment between simulation and  
 123 the real camera. A new soccer skill is specified by choosing its task objects, task variables, rewards,  
 124 curricula, and optional contact conditioning. Once these elements define what the robot should ac-  
 125 complish, the same training pipeline can optimize a policy under the same deployment constraints.

### 126 3.3 Visibility-Aware Auxiliary Geometry

127 Learning task geometry from depth only through task returns is  
 128 inefficient: task objects can occupy a small image region, while  
 129 the gradient from PPO returns must pass through the actor and  
 130 the depth encoder before shaping visual features. We therefore  
 131 attach a visibility-aware auxiliary geometry module to the depth  
 132 latent and supervise it with privileged simulation labels during  
 133 training. As shown in Fig. 3, the module predicts object-level  
 134 geometry and visibility inside the policy, so the actor receives  
 135 direct object-level supervision without adding a separate run-  
 136 time perception module.

137 Let  $\mathbf{z}_t^D = f_D(\mathbf{D}_t)$  be the depth latent. For a single task  
 138 object  $\text{obj}_1$ , the module predicts local geometry and visibil-  
 139 ity as  $(\hat{\mathbf{y}}_t^1, \hat{\mathbf{q}}_t^1) = h_{\text{aux}}^1(\mathbf{z}_t^D)$ , where  $\mathbf{y}_t^1$  is the task-defined ob-  
 140 ject geometry in the robot frame. The same construction ex-  
 141 tends to  $\mathcal{O} = \{\text{obj}_i\}_{i=1}^n$  by assigning each object an indexed  
 142 prediction  $(\hat{\mathbf{y}}_t^i, \hat{\mathbf{q}}_t^i)$ , target geometry  $\mathbf{y}_t^i$ , and visibility  $\mathbf{v}_t^i$ .  
 143 The actor-side representation is the concatenated gated feature

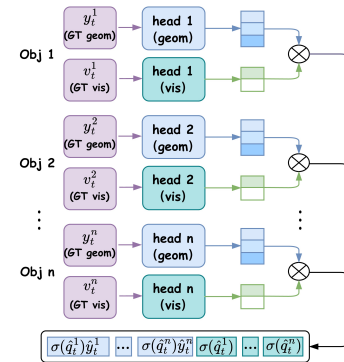


Figure 3: Scalable auxiliary ge-  
 ometry. Each task object is as-  
 signed a geometry head and a vis-  
 ibility head supervised by privi-  
 leged labels during training; their  
 outputs are visibility-gated and  
 concatenated into the actor fea-  
 ture, enabling the same mecha-  
 nism to extend from one object to  
 $n$  objects.

144  $\hat{\mathbf{u}}_t = [\sigma(\hat{q}_t^1)\hat{\mathbf{y}}_t^1, \dots, \sigma(\hat{q}_t^n)\hat{\mathbf{y}}_t^n, \sigma(\hat{q}_t^1), \dots, \sigma(\hat{q}_t^n)]$ . Thus adding  
 145 a task-relevant object only adds another indexed geometry / visibility head and mask; the depth input,  
 146 actor interface, and 25-DoF control loop remain unchanged.

147 Visibility is learned jointly with geometry because a head-mounted camera can temporarily lose any  
 148 task object. We supervise coordinates only when the object is visible and train visibility confidence  
 149 with a shared binary objective, giving  $\mathcal{L}_{\text{aux}} = \sum_{i=1}^n \lambda_i v_t^i \ell_i(\hat{\mathbf{y}}_t^i, \mathbf{y}_t^i) + \sum_{i=1}^n \lambda_{v_i} \text{BCE}(\hat{q}_t^i, v_t^i)$ . Details  
 150 are provided in Appendix C. To avoid forcing the actor to rely on an untrained predictor early in  
 151 learning, we blend ground-truth and predicted features as  $\mathbf{u}_{t,k}^{\text{actor}} = \alpha_k \mathbf{u}_t^{\text{gt}} + (1 - \alpha_k) \hat{\mathbf{u}}_t$ , where  $\alpha_k$   
 152 is annealed from  $\alpha_{\text{start}}$  to  $\alpha_{\text{end}}$  over  $K_{\text{anneal}}$  policy updates. Privileged geometry therefore serves as  
 153 a representation-learning curriculum, while the final deployed actor consumes only depth-predicted  
 154 geometry.

### 155 3.4 Skill Instantiation

156 Once the deployment interface in Sec. 3.2 is fixed, a soccer skill is specified by a task definition  
 157 rather than by switching runtime modules. We denote such a definition by  $\mathcal{T} = (\mathcal{O}, \mathbf{c}_t, \mathbf{y}_t, r_t, \mathcal{K})$ ,  
 158 where  $\mathcal{O}$  is the set of task objects,  $\mathbf{c}_t$  is the optional command,  $\mathbf{y}_t$  is the task geometry supervised  
 159 through Sec. 3.3,  $r_t$  defines the contact objective, and  $\mathcal{K}$  is the curriculum. Below we describe two  
 160 representative contact regimes under this task-definition view; the detailed reward terms, curriculum  
 161 stages, and parameter ranges are deferred to Appendix D.1 and Appendix D.2.

#### 162 3.4.1 Goal-Directed Shooting

163 Shooting is defined by the ball-goal geometry. Let  $\mathbf{b}_t$  and  $\mathbf{g}_t$  denote the ball position and goal center  
 164 in the robot frame, and let  $\mathbf{d}_t^{bg} = \frac{\mathbf{g}_t - \mathbf{b}_t}{\|\mathbf{g}_t - \mathbf{b}_t\|_2 + \epsilon}$  be the normalized ball-to-goal direction. A behind-ball  
 165 target can then be written as  $\mathbf{x}_t^{\text{behind}} = \mathbf{b}_t - \rho \mathbf{d}_t^{bg}$ , which turns chasing, alignment, and contact into  
 166 one geometric target family. In this task, the object set is  $\mathcal{O}_{\text{shoot}} = \{\text{ball}, \text{goal}\}$ , the task geometry  
 167 is  $\mathbf{y}_t^{\text{shoot}} = (\mathbf{b}_t, \mathbf{g}_t)$ , and the objective is to strike the ball so that its exit motion moves toward  
 168 the goal. The reward is organized around run-up, contact, terminal success, visual maintenance,  
 169 and safety regularization, while the curriculum gradually broadens the initial ball placement and  
 170 approach difficulty. Active vision is maintained through the head DoFs: the head keeps the ball  
 171 within view, and the body gradually follows the viewing direction. Detailed reward terms, reset  
 172 ranges, and stage boundaries are given in Appendix D.1.

#### 173 3.4.2 Directional Dribbling

174 Dribbling is defined by a direction command and sustained ball regulation. Let  $c_t$  be the com-  
 175 manded direction angle and  $\mathbf{d}_t^{\text{cmd}} = [\cos c_t, \sin c_t]$  the corresponding horizontal direction. The  
 176 task object set is  $\mathcal{O}_{\text{dribble}} = \{\text{ball}\}$ , the command is  $\mathbf{c}_t = c_t$ , and the task geometry can be writ-  
 177 ten as  $\mathbf{y}_t^{\text{dribble}} = (\mathbf{b}_t, \mathbf{d}_t^{\text{cmd}}, d_t)$ , where  $d_t$  is the robot-ball horizontal distance. Unlike shoot-  
 178 ing, the objective is not a single high-speed strike; the policy must keep the ball within a con-  
 179 trollable band and generate progress along  $\mathbf{d}_t^{\text{cmd}}$ . We express the controllable-band condition by  
 180  $\chi_t^{\text{band}} = \mathbb{I}(|d_t - d_0| < \Delta_d, v_{lo} < \|\mathbf{v}_t^b\| < v_{hi}, \mathbf{v}_t^b \cdot \mathbf{d}_t^{\text{cmd}} > 0)$ . The reward covers commanded  
 181 progress, re-approach, heading, visual maintenance, and out-of-band penalties. Continuous drib-  
 182 bling further uses contact-intent-conditioned AMP and phase masks so approach, touch, follow-  
 183 through, and recovery emphasize different motion states. The exact command set, controllable-band  
 184 parameters, contact-intent construction, and curriculum rules are listed in Appendix D.2.

### 185 3.5 Training and Deployment Pipeline

186 The deployment policy uses a single-frame head-mounted depth image, a five-step proprioceptive  
 187 history, and task variables to output 25-DoF joint PD targets. Since this depth image is the main sim-  
 188 to-real interface, we align simulation and deployment before it enters the actor. As shown in Fig. 4,  
 189 a raw frame  $\mathbf{D}_t$  is processed to  $\tilde{\mathbf{D}}_t$ . During training, the aligned frame is further perturbed to reduce  
 190 dependence on clean depth and serve as domain randomization for sim-to-real transfer [44, 45]. The  
 191 remaining training and deployment details, including PPO/AMP/HIM [46] architecture, training  
 192 and deployment details are provided in Appendices A and E.

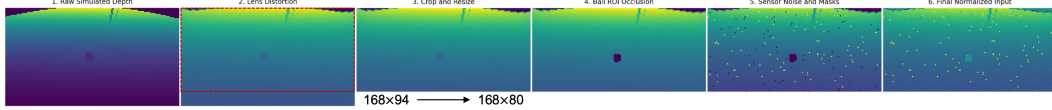


Figure 4: Depth alignment transforms the raw depth frame into the aligned policy input through distortion correction, cropping from  $168 \times 94$  to  $168 \times 80$ , ball ROI occlusion, sensor noise, random masks and normalization, so the same single-frame depth interface can be used both in sim and real.

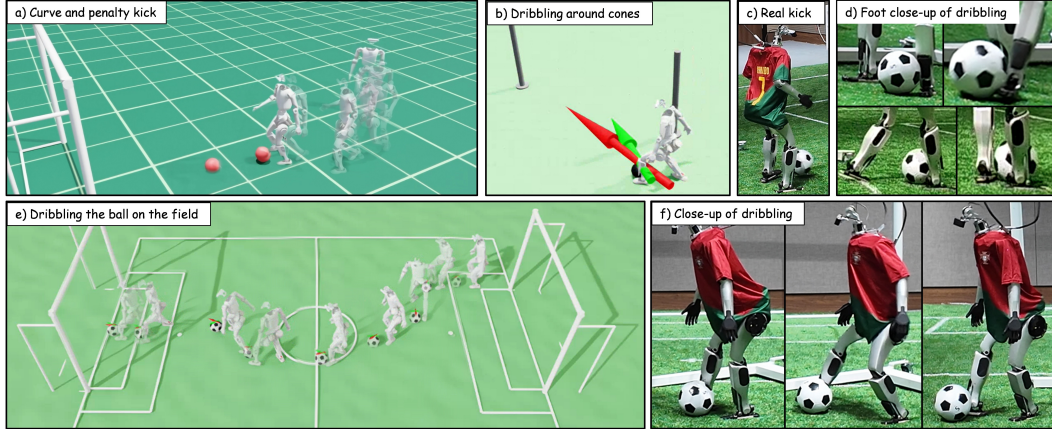


Figure 5: Qualitative examples of learned soccer contact skills in simulation and on the real robot. The policy performs directional dribbling on a field, cone-dribbling in simulation, goal-directed free kick, real-world close-up dribbling with sustained foot-ball regulation, and real-world kicking.

## 193 4 Experiments

194 We evaluate whether **DAVIS** supports both simulated and real-world soccer contact skills, as illus-  
 195 trated in Fig. 5. And we answer the following questions: **Q1**. Can a depth-only end-to-end policy  
 196 perform soccer contact skills? **Q2**. Which training structures are necessary? **Q3**. Does the learned  
 197 policy transfer to the real E1 robot?

### 198 4.1 Simulation Experiments

199 To answer Q1, we extensively evaluate **DAVIS** in IsaacLab simulator on two representative skill  
 200 categories in soccer: shooting and dribbling.

201 **Shooting**: The robot is required to perform shooting with randomly initialized robot or ball posi-  
 202 tions. In the penalty kick task, the robot starts within angle of  $[-100^\circ, 100^\circ]$  behind the ball and  
 203 distances of 2–3.5 m. In the free kick task, the robot is initialized 5 m in front of the goal, and is  
 204 required to kick the ball within the region  $x \in [0, 4]$  m and  $y \in [-3, +3]$  m. The difficulty level  
 205 categories are shown in Fig. 6. We record the success rate of the task. Experimental results in Ta-  
 206 ble 1 demonstrating strong positional generalization capability. By analyzing the kinematic pattern  
 207 during the shooting process in Fig. 6, we observe that the robot exhibits a human-like gait pattern.

208 **Dribbling**: The robot is required to complete a repeated-S slalom task.  $N$  is the number of poles,  
 209 and  $\alpha$  is the turning angle of instruction. A task is considered successful if the robot passes through  
 210 all waypoints in the predefined order. To ensure consistency, we use a fixed instruction sequence for  
 211 all experiments. The policy achieves an average success rate of 0.65 across 900 trials. Experimental  
 212 results in Table 2 demonstrate that more than half of the task configurations obtain success rates  
 213 above 0.65, showing robustness in long trajectory and extreme angles. Visualization in Fig. 7 shows  
 214 the dynamics of the policy.

215 Experiments show that **DAVIS** achieves high performance in all tasks. These results demonstrate  
 216 that our framework are able to handle highly dynamic locomotion and fine-grained soccer manipu-  
 217 lation.

| Task         | Success rate (%) |        |       |       |
|--------------|------------------|--------|-------|-------|
|              | Easy             | Medium | Hard  | Total |
| Penalty kick | 94.80            | 95.80  | 55.50 | 84.87 |
| Free kick    | 100.00           | 88.00  | 71.00 | 85.36 |

Table 1: Shooting simulation results

| $N$                    | 3           |             |      | 5           |             |      | 7           |      |      |
|------------------------|-------------|-------------|------|-------------|-------------|------|-------------|------|------|
| $\alpha$ ( $^\circ$ )  | 60          | 90          | 120  | 60          | 90          | 120  | 60          | 90   | 120  |
| SR $\uparrow$          | 0.71        | <b>0.77</b> | 0.65 | 0.66        | <b>0.72</b> | 0.55 | <b>0.68</b> | 0.64 | 0.43 |
| $E_{vel}$ $\downarrow$ | <b>0.69</b> | 0.74        | 0.91 | 0.74        | <b>0.73</b> | 0.80 | <b>0.71</b> | 0.81 | 0.97 |
| $R_{bib}$ $\uparrow$   | 0.73        | <b>0.74</b> | 0.64 | <b>0.76</b> | 0.75        | 0.70 | <b>0.76</b> | 0.72 | 0.60 |

Table 2: Dribbling simulation results.

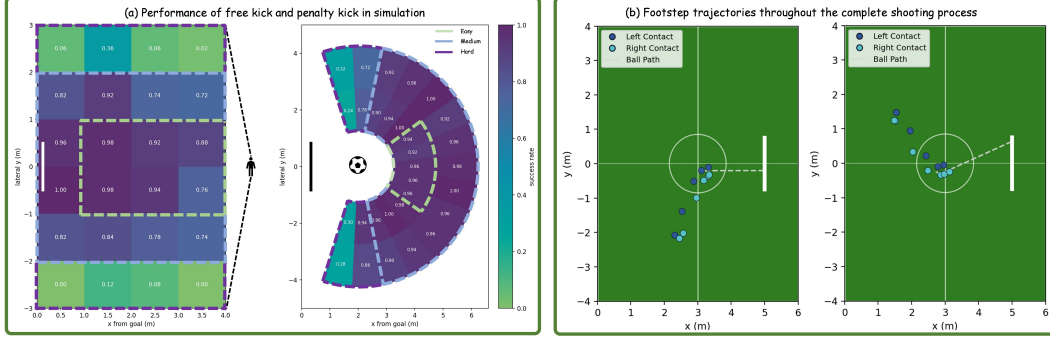


Figure 6: (a) Free-kick and Penalty-kick performance in simulation. Each region uses over 50 trials. Darker colors indicate higher success rates, and dashed lines mark difficulty regions. (b) Footstep trajectories during shooting. The policy shows human-like gait and speed pattern.

## 4.2 Ablation Study

To verify the necessity of key structures and answer Q2, we conduct extensive ablation studies on three tasks in simulation and record different metrics in Table 3. All methods share the same training parameters and evaluation settings.

**Ablation on curriculum.** We remove the ramp-up and scaling curriculum. All reward terms are activated with full weights from the beginning. Without the progressive curriculum, the policy requires more training iterations to converge, and performance is significantly worse than baseline.

**Ablation on auxiliary head.** Our policy incorporates a dedicated object position prediction head. To validate effectiveness, we mask the predicted positions fed into the actor, while retaining the network architecture and input dimension. In addition, we disable the aux loss and stop backpropagation. As expected, without the explicit predictions, model struggles to localize objects to make contact and success drops to zero.

**Ablation on active vision.** Active vision is another key design. We fix the 2-DoF head joints and replace the original head-based rewards with body alignment rewards and corresponding curriculum is adjusted. After fixing head joints, the ball frequently moves out FOV or enters blind spots, causing collapse. Another worth-noting observations is that robot is unable to locate the gate. The performance drops near to zero and with a high possibility to fall.

## 4.3 Real-World Deployment

To answer Q3, we deploy our policies on the Noetix E1 humanoid robot equipped with a ZED2i depth camera on 2-DoF head. For the shooting experiments, we select 12 representative positions according to levels used in simulation. For dribbling experiments, we design three tasks including reaching target position, obstacle traversal, and slalom dribbling. To better reflect real-world soccer scenarios, all experiments are conducted on a grass surface. Fig. 1 and Fig. 5 show that our policy could perform soccer task with high dynamics, precise actions. Quantitative results in Table 5 and Table 4 demonstrate that our policy maintains a high success rate in real deployment. Detailed experimental settings and implementation details are provided in the Appendix B.

| Metric                      | Easy  | Medium | Hard  |
|-----------------------------|-------|--------|-------|
| Success rate (%) $\uparrow$ | 68.42 | 57.14  | 55.00 |

Table 4: Shooting real world experiment results.

| Metric                      | Easy  | Medium | Hard  |
|-----------------------------|-------|--------|-------|
| Success rate (%) $\uparrow$ | 78.57 | 67.86  | 60.71 |

Table 5: Dribbling real world experiment results.

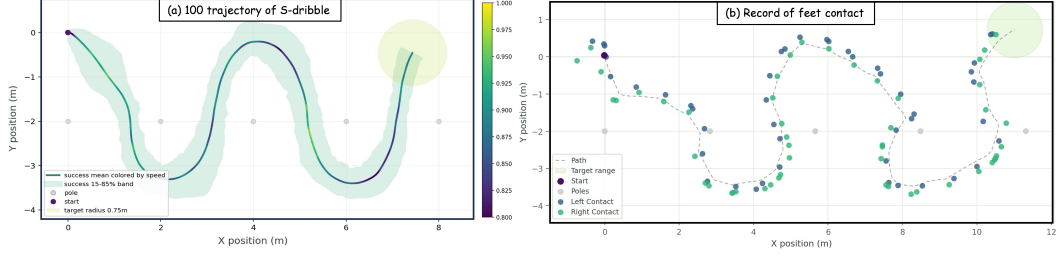


Figure 7: Performance of repeated S-slalom in simulation. (a) Statistical results of 100 S-slalom trajectories. (b) Foot-contact record from a representative successful trajectory.

| Methods            | Penalty Kick  |                        |                          | Free Kick     |                        |                          | Repeated-S Slalom |                        |                               |
|--------------------|---------------|------------------------|--------------------------|---------------|------------------------|--------------------------|-------------------|------------------------|-------------------------------|
|                    | SR $\uparrow$ | $E_{ang}$ $\downarrow$ | $r_{contact}$ $\uparrow$ | SR $\uparrow$ | $E_{ang}$ $\downarrow$ | $r_{contact}$ $\uparrow$ | SR $\uparrow$     | $E_{vel}$ $\downarrow$ | $r_{ball-in-band}$ $\uparrow$ |
| Full method        | <b>0.84</b>   | <b>23.29</b>           | <b>99.75</b>             | <b>0.80</b>   | <b>30.18</b>           | <b>99.27</b>             | <b>0.70</b>       | 0.79                   | <b>0.74</b>                   |
| w/o curriculum     | 0.62          | 27.40                  | 32.53                    | 0.55          | 34.42                  | 87.67                    | 0.51              | <b>0.77</b>            | 0.69                          |
| w/o auxiliary head | 0             | —                      | 0                        | 0             | —                      | 0                        | 0                 | 1.45                   | 0.05                          |
| w/o active vision  | 0             | —                      | 0                        | 0.36          | 97.81                  | 1.71                     | 0                 | 1.25                   | 0.03                          |

Table 3: Simulation ablations. **Penalty/ Free-kick** results use over 900 shooting trajectories, and **Repeated S-slalom** uses over 100 dribbling trajectories. Metrics are **SR** (success rate),  $E_{ang}$  (angular error between the robot body heading and the robot-to-goal direction),  $r_{contact}$  (successful ball-contact ratio),  $E_{vel} = \|\mathbf{v}_{cmd} - \mathbf{v}_{ball}\|_2$ , and  $r_{ball-in-band}$  (time in the control band). **The full method performs best overall**; See Appendices B and F for more details.

## 5 Conclusion

We presented a depth-only end-to-end active-vision framework **DAVIS** for humanoid soccer skills. The main idea is simple: keep the deployment interface minimal, but place enough structure inside training so that the policy can learn soccer-specific geometry, visibility, and whole-body contact behavior from depth and proprioception alone.

Within the same framework, we instantiated goal-directed shooting and directional dribbling. The two skills share the same observation and control interface, but differ in task geometry, reward shaping, curriculum progression, and motion-prior conditioning. This shows that a single depth-to-control framework can support more than one contact-rich soccer behavior without introducing a separate perception-planning stack at runtime.

Our experiments evaluate simulation performance, real-robot transfer, and contribution of each key component. The overall takeaway is that active head motion, visibility-aware auxiliary geometry, curriculum learning, and AMP-style motion regularization are most useful when they are treated as training structure for a single deployed policy rather than as separate runtime modules.

## 6 Limitations and Future Work

Two limitations remain. The first is perceptual: depth-only sensing keeps the interface compact but ties the policy to a single head-mounted camera, inheriting its occlusions, dropouts, and imperfectly simulated depth—a residual sim-to-real gap that limits real-world accuracy. The second sits upstream: the task-definition interface unifies shooting and dribbling, yet each skill needs hand-tuned rewards, curricula, and contact conditioning, leaving the pipeline a recipe, not an automatic one.

Closing the perceptual gap is largely engineering—sensor-noise modeling and real-world fine-tuning. Automating skill construction is the more open problem, where language-conditioned specification and automatic curricula could generate new behaviors on the same interface.

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## Appendix

### A Real Robot System Setup

Both soccer skills are deployed on the Noetix E1 using a shared onboard control stack. The exported ONNX actor maps a single aligned head-mounted depth frame, proprioceptive history, and an optional low-dimensional task command to 25-DoF joint-position targets. The policy runs at 50 Hz, while the low-level PD controller runs at 200 Hz and the depth stream is updated at 15 Hz. No external localization, off-board state estimator, object detector, or runtime soccer planner is used during real-robot trials.

Policies are trained in Isaac Lab (built on Isaac Sim 5.0.0) on eight NVIDIA RTX 4090D GPUs for 20,000 PPO iterations before ONNX export. The 25-dimensional action vector is kept unchanged across training, export, and deployment.

### B Simulation and Real-World Evaluation Protocols

Table 6 defines the metrics used by the simulation, ablation, and real-world result tables. The same definitions are used in the main paper and appendix tables so that success rates are comparable across tasks and deployment settings.

Table 6: Metric definitions used throughout the evaluation.

| Metric                                       | Definition                                                                               |
|----------------------------------------------|------------------------------------------------------------------------------------------|
| Success rate (SR)                            | Scoring for shooting, and passing all waypoints for dribbling.                           |
| Valid contact ratio ( $r_{\text{contact}}$ ) | Fraction of shooting trials with a valid foot-ball contact and goal-directed ball exit.  |
| Angular error ( $E_{\text{ang}}$ )           | Heading error between the robot body and the robot-to-goal direction.                    |
| Velocity error ( $E_{\text{vel}}$ )          | Mean error between commanded and actual ball velocity during dribbling.                  |
| Ball-in-band ratio ( $R_{\text{bib}}$ )      | Fraction of engaged dribbling time in which the ball stays inside the controllable band. |

Each main-paper result table follows either Table 6 or the task-specific protocol described in this appendix. When a metric is computed differently for simulation and real-world deployment, the difference is reported with the corresponding result table.

#### B.1 Directional Shooting Simulation Protocol

The shooting simulation results in Table 1 and Fig. 6 are evaluated on two tasks: penalty kick and free kick. In the free-kick task, the robot starts from a fixed position 5 m in front of the goal center and faces the goal, while the ball is randomly placed within  $x \in [0, 4]$  m and  $y \in [-3, +3]$  m. In the penalty-kick task, the ball is fixed 1.5 m in front of the goal center, while the robot is initialized behind the ball within an angular range of  $[-100^\circ, 100^\circ]$  and a distance range of 2–3.5 m, facing the ball. The difficulty level diagram of the shooting task is shown as Figure 8.

To evaluate spatial generalization, we use a stratified random initialization scheme. The penalty-kick sector is divided into  $2 \times 15$  regions, and the free-kick ball-placement area is discretized into  $1 \text{ m} \times 1 \text{ m}$  grid cells. We sample the same number of trials from each region to reduce sampling bias and ensure uniform coverage.

We report SR,  $E_{\text{ang}}$ , and  $r_{\text{contact}}$  following the definitions in Table 6. The angular error is computed as

$$E_{\text{ang}} = \arccos \left( \frac{\mathbf{v}_1^\top \mathbf{v}_2}{\|\mathbf{v}_1\| \|\mathbf{v}_2\|} \right) \quad (1)$$

where  $\mathbf{v}_1$  is the robot body-heading vector and  $\mathbf{v}_2$  points from the robot base to the goal center. All shooting simulation evaluations use the same policy checkpoint, IsaacLab environment configuration, 5 m depth range, and depth preprocessing pipeline described in Sec. 3.5.

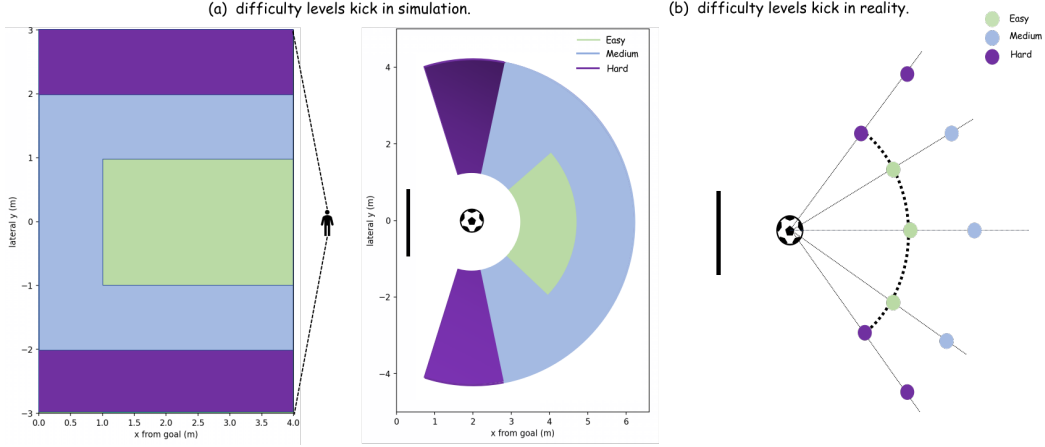


Figure 8: Multi-position Shooting Simulation and Real Machine Difficulty Level Classification

## B.2 Directional Dribbling Simulation Protocol

The dribbling simulation results in Table 2 are evaluated by replaying a fixed policy checkpoint in the same IsaacLab environment and control configuration used for training. The benchmark uses a parametrized repeated-S slalom fixture (Table 7) over  $N \in 3, 5, 7$  trajectory turns and  $\alpha \in 60, 90, 120^\circ$  turning angles, resulting in nine evaluation tasks and 900 trials in total. The ball is initialized 0.6 m in front of the robot, while the robot pose and AMP reference-state reset vary across rounds.

Table 7: Directional dribbling simulation fixture.

| Field            | Definition                                                           | Value                                                     |
|------------------|----------------------------------------------------------------------|-----------------------------------------------------------|
| Task grid        | Repeated-S slalom with trajectory turns $N$ and turn angle $\alpha$  | $N \in 3, 5, 7, \alpha \in 60, 90, 120^\circ$             |
| Waypoints        | $N+2$ turning waypoints plus $N+1$ midline gate points               | $2N+3$ total                                              |
| Geometry         | Fixed arm length for waypoint generation                             | $L = 4.0$ m                                               |
| Command          | Waypoint attraction and pole repulsion, snapped to 12 direction bins | $k_{\text{att}} = 1.0, k_{\text{rep}} = 0.6, d_0 = 1.2$ m |
| Ball spawn       | Fixed point directly in front of the robot                           | 0.6 m                                                     |
| Engagement       | Metrics start after first ball contact                               | threshold 0.15 m/s; timeout 5 s                           |
| Waypoint arrival | Radius for advancing to the next waypoint                            | 0.75 m                                                    |
| Segment timer    | Time budget refreshed after each waypoint arrival                    | 5 s                                                       |
| Failure cases    | Fall, timeout, or ball leaving the controllable band                 | —                                                         |

Following Table 6, we report SR,  $E_{\text{vel}}$ , and  $R_{\text{bib}}$  for dribbling. All metrics are computed over the post-engagement window, i.e., after the robot first takes over the ball, so the initial approach phase is excluded. Specifically,  $E_{\text{vel}}$  is the mean L2 error between the commanded ball velocity and the measured ball velocity over the engaged steps, where the commanded velocity points along

the current dribbling command with a peak speed of 1.0 m/s.  $R_{\text{bib}}$  is accumulated over the same engaged window and is undefined for rounds that never enter the controllable band.

The in-band predicate follows the deployed termination settings: the robot–ball distance must lie within 0.10 m of the 0.42 m target, the ball speed must lie in  $[0.20, 1.50]$  m/s, and the ball velocity must not point against the command direction. A round fails if the ball remains out of band for more than 3.5 s or exceeds the 0.80 m far-field scope. For fair aggregation, every policy is evaluated with the same checkpoint selection rule, fixed ball spawn, slalom grid, band thresholds, and reset distribution; only the policy weights differ across ablations.

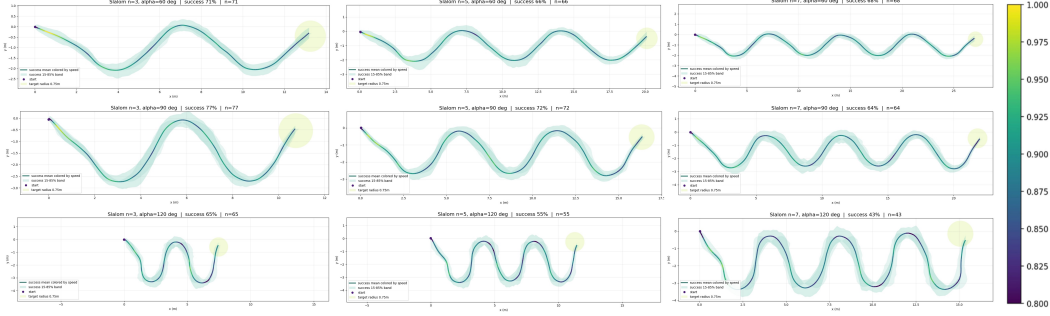


Figure 9: Trajectory visualization for the repeated-S slalom dribbling benchmark. Each subplot corresponds to one task configuration  $(N, \alpha)$ , where  $N$  is the number of turns and  $\alpha$  is the turning angle. The colored curve shows the mean successful ball trajectory with speed-based coloring, and the shaded region indicates the 15–85% success band. The start point and target region are marked for reference. The title of each subplot reports the success rate and the number of successful rounds.

446

**Trajectory visualization.** Figure 9 visualizes the dribbling trajectories under the repeated-S slalom benchmark. Each panel corresponds to one task configuration defined by the number of turns  $N$  and turning angle  $\alpha$ . For each configuration, we plot the engagement-anchored ball trajectories in a local forward-left frame, together with the mean successful trajectory and its success band. Trajectories are color-coded by ball speed, while the start point and target region are marked for reference. These plots provide a qualitative view of how the policy regulates the ball across different slalom difficulties, and use the same checkpoint and evaluation protocol as the metric tables.

### 454 B.3 Ablation Experiment Protocol

For all variant in ablation experiments, We follow the identical test settings, evaluation protocols and metrics mentioned in Appendix B.1 and B.2. Meanwhile, we adopt identical learning hyperparameters, training steps and parallel environments as those used in simulation experiments to guarantee consistent experimental variables. For safety concerns, no ablation studies are conducted on real-world robots.

**Ablation on curriculum.** The ramp-up of head and body rewards groups is removed. From step 0 onward, body and head reward scale are fixed to 1. No staged scaling is applied to ball-chasing and gaze-related terms. All task rewards and ball-chasing components take full effect from the beginning of training, eliminating the staged curriculum of “movement first, head activation later”.

**Ablation on auxiliary head.** In the full architecture, depth maps are encoded into latent features via a CNN network. Three prediction heads (goal, ball, visibility) then generate geometric and visibility features, which are concatenated to the actor input. An auxiliary loss is constructed using privileged ground truth to update the encoder and prediction heads. In this ablation, we retain the network structure of the prediction heads and the dimension of the auxiliary input to the actor, but mask all auxiliary features fed into the actor to 0, disable the auxiliary loss, and cut off its backpropagation. This renders the prediction head fully non-functional during both training and inference without changing the overall model structure.

472 **Ablation active vision.** The reward structure for ball chasing remains unchanged, but the control  
 473 objective is switched from “where to orient the head” to “where to rotate the body”. The two original  
 474 terms relying on head joint angles are replaced with body yaw/pitch to align with the ball and target  
 475 line. Accordingly, the curriculum learning is adjusted from head reward scale to body reward scale,  
 476 allowing the constraint to be gradually activated together with the body-following reward, instead  
 477 of using an independent head-oriented curriculum. Figure

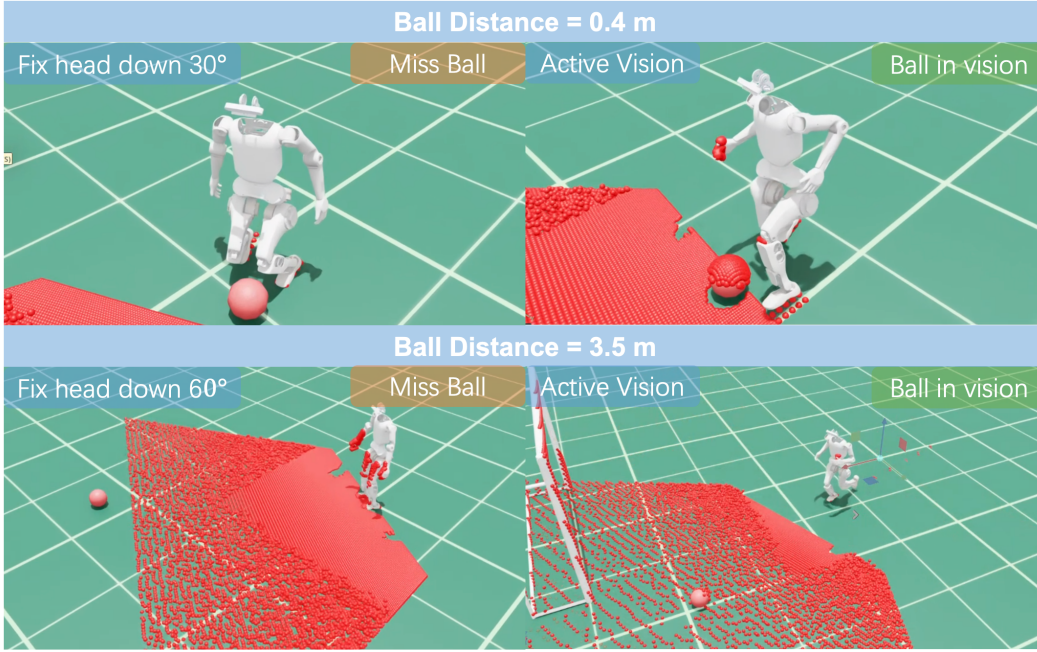


Figure 10: Ablation on active vision.

#### 478 B.4 Real World Experiment Protocol

479 All real-world experiments are conducted according to the experimental settings detailed in Ap-  
 480 pendix A. All real-world experiments are conducted with safety protection and on a real grass field.  
 481 In real-world evaluations, we report only the task success rate defined in Table 6.

482 **Real World Shooting Experiment.** For the real-world shooting experiments showed in Table 4,  
 483 we adopt the penalty-kick setting. The ball is placed 1.5 m in front of the goal, and the robot per-  
 484 forms penalty kicks from different initial positions. The goal dimensions match those in simulation:  
 485 half-width 1.20 m, height 1.80 m, and depth 1.00 m. The shooting experiments are conducted by  
 486 replaying a fixed policy checkpoint in the same IsaacLab environment and control setup used during  
 487 training and simulation testing.

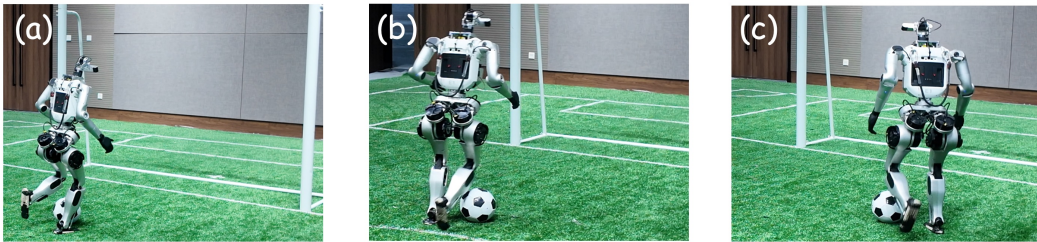


Figure 11: Multi-directional shooting real-world example

488 To ensure experimental consistency, we evaluate the policy from 12 predefined positions within  
 489 the test area. The Easy, Medium, and Hard settings were categorized according to both the target

Table 8: Target point configurations used in the real-world evaluation.

| Difficulty | Distance (m) | Target Angles                             |
|------------|--------------|-------------------------------------------|
| Easy       | 2.5          | $-30^\circ, -5^\circ, +5^\circ +30^\circ$ |
| Medium     | 3.0          | $-30^\circ, -5^\circ, +5^\circ +30^\circ$ |
| Hard       | 2.5–3.0      | $-45^\circ, +45^\circ$                    |

distance and the required heading change, as shown in Table 8. In all experiments, the robot was initialized facing the goal. A successful trial required the robot to score by kicking the ball into the goal with a single kick attempt. Score is defined as the ball completely crossing the goal line.

**Real World Dribbling Experiment.** For the real-world dribbling experiments showed in Table 5, we design three experimental settings with increasing difficulty. The Easy difficulty level corresponds to the **Reaching Target Position** task. In this task, the robot was required to dribble the ball to a designated target location. The target was randomly generated for each trial at a distance of 4–5 m from the robot and within an angular range of  $\pm 30^\circ$  relative to its initial heading. The Medium difficulty level corresponds to the **Obstacle Traversal** task. In this experiment, the robot was required to dribble the ball to a designated target location while avoiding a predefined obstacle. The obstacle was positioned 3–4 m away from the robot, and the target point was typically located approximately 3 m beyond the obstacle. Depending on the obstacle configuration, successful traversal required the robot to execute a heading change of approximately  $60^\circ$ – $120^\circ$  during obstacle avoidance. The hard difficulty level corresponds to the **Slalom Dribbling** task. In this experiment, the robot was required to complete a continuous dribbling task by navigating through a sequence of target locations. Consecutive target points were separated by 2–3 m, and transitioning between targets required a heading change of approximately  $60^\circ$ – $120^\circ$ , consistent with the previous experimental setting. For each trial, the robot was required to visit 2–4 target points sequentially while maintaining control of the ball. For all experiments described above, the ball spawn distance was fixed at 0.6 m.

For each of the three experimental settings, a trial was counted as successful when the robot remained upright throughout the execution and both the ball and the human were simultaneously located within a 1 m radius of the target point. Similarly, the real-world experiments used the identical setup and policy checkpoint employed during simulation training and evaluation. To ensure consistency, all trials were performed by the same operator using a joystick controller.

## C Observation Space and Policy Architecture

This section provides implementation details for the deployable observation interface, policy architecture, PPO / AMP-HIM optimization, and the auxiliary geometry and visibility losses used by DAVIS.

### C.1 Observation Space and Policy Architecture

The actor input consists of a single aligned depth frame, a 5-step proprioceptive history, and an optional low-dimensional task command. The shooting policy uses an 81-D proprioceptive vector, while the dribbling policy uses an 84-D proprioceptive vector. The proprioceptive vector includes base angular velocity, projected gravity, relative joint positions, relative joint velocities, and the previous action. The depth image is encoded by a `LightDepthEncoder` into a 32-D latent feature. Both the actor and critic MLPs use hidden layers [512, 256, 128] with ELU activations and empirical observation normalization.

During training, the critic additionally receives privileged quantities, including base linear velocity, foot contact states, terrain height scan, ball and goal geometry, auxiliary labels, and curriculum variables. These signals are used only for training-time value learning, reward computation, or auxiliary supervision, and are not actor inputs at deployment. The HIM estimator uses the proprioceptive

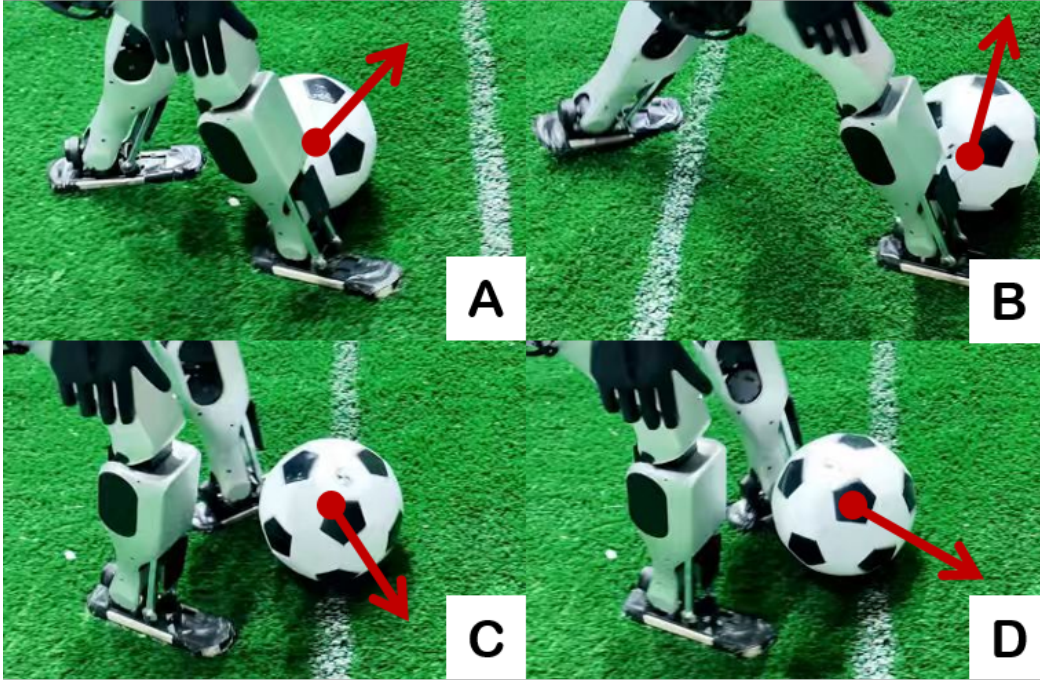


Figure 12: Foot snapshot in dribbling experiment.

531 history to produce a 3-D latent estimate of hidden robot state, which is concatenated with the actor  
 532 features.

533 The depth latent is supervised by auxiliary geometry and visibility heads. The geometry heads  
 534 predict local 3-D positions of the ball and goal, and the visibility head predicts whether each object  
 535 is visible in the current depth frame. Ground-truth geometry and visibility are linearly replaced by  
 536 predicted geometry.

Table 9: Observation and network configuration.

| Item                      | Configuration                                            |
|---------------------------|----------------------------------------------------------|
| Actor sensory input       | Single-frame depth image + 5-step proprioceptive history |
| Optional command          | Low-dimensional task command if enabled                  |
| Shooting proprioception   | 81-D                                                     |
| Dribbling proprioception  | 84-D                                                     |
| Depth encoder             | LightDepthEncoder                                        |
| Depth latent              | 32-D                                                     |
| HIM estimate output       | 3-D                                                      |
| Actor MLP                 | [512, 256, 128], ELU                                     |
| Critic MLP                | [512, 256, 128], ELU                                     |
| Actor normalization       | Enabled                                                  |
| Critic normalization      | Enabled                                                  |
| Auxiliary geometry heads  | Goal 3-D, ball 3-D                                       |
| Auxiliary visibility head | 2-D: goal visible, ball visible                          |

537 The AMP discriminator uses a 136-D single-frame state composed of joint position, key-body po-  
 538 sitions, base linear velocity, base angular velocity, joint velocity, and foot-contact masks. We mask  
 539 the head-related dimensions in the discriminator input because head yaw and pitch are optimized for  
 540 active visual tracking rather than locomotion-style imitation.

Table 10: PPO and AMP-HIM optimization settings.

| Item                        | Value                                  |
|-----------------------------|----------------------------------------|
| Rollout length              | 24 steps per environment               |
| PPO epochs                  | 5                                      |
| Mini-batches                | 4                                      |
| Learning rate               | $1 \times 10^{-3}$ , adaptive schedule |
| Discount factor             | $\gamma = 0.99$                        |
| GAE parameter               | $\lambda = 0.95$                       |
| PPO clip                    | 0.2                                    |
| Entropy coefficient         | 0.01                                   |
| Value loss coefficient      | 1.0                                    |
| Max gradient norm           | 1.0                                    |
| Discriminator LR            | $5 \times 10^{-6}$                     |
| Discriminator loss          | Wasserstein loss                       |
| Discriminator hidden layers | [1024, 512]                            |
| AMP replay buffer           | 200000                                 |
| AMP reward coefficient      | 0.8                                    |
| AMP reward interpolation    | 0.8                                    |
| Auxiliary geometry loss     | goal: 1.0, ball: 1.0                   |
| Visibility loss             | goal: 0.2, ball: 0.2                   |

## 541 C.2 Auxiliary Geometry and Visibility Losses

542 Let  $z_t^d$  denote the 32-D depth latent produced by the `LightDepthEncoder`. For each task object  $i$   
543 (the ball, and the goal in the shooting task), an auxiliary head predicts a relative object geometry  $\hat{\mathbf{y}}_t^i$   
544 and a visibility logit  $\hat{v}_t^i$  from the shared latent,

$$(\hat{\mathbf{y}}_t^i, \hat{v}_t^i) = h_{\text{aux}}^i(z_t^d), \quad p_t^i = \sigma(\hat{v}_t^i). \quad (2)$$

545 The geometry is expressed in the robot local frame, and its range component is passed through  
546 a softplus so that predicted distances stay non-negative and remain consistent with the supervised  
547 target and the deployed predictor.

548 Simulator labels provide a target geometry  $\mathbf{y}_t^{i*}$  and a binary visibility label  $v_t^{i*}$ . The latter is a  
549 *trust* predicate rather than a raw field-of-view test,  $v_t^{i*} = \mathbb{I}[\text{object in FOV} \wedge \rho_t^{i*} < \rho_{\max} - \delta]$ , so  
550 it marks whether the prediction is reliable at the current range  $\rho_t^{i*}$ . Geometry is supervised only  
551 on visible frames and normalized by the visible-sample count, with a wrap-safe angular term and a  
552 range term, while visibility is trained by a binary cross-entropy on the logits:

$$\mathcal{L}_{\text{geo}} = \sum_i v_t^{i*} \ell(\hat{\mathbf{y}}_t^i, \mathbf{y}_t^{i*}), \quad \mathcal{L}_{\text{vis}} = \sum_i \lambda_v \text{BCEWithLogits}(\hat{v}_t^i, v_t^{i*}), \quad (3)$$

$$\ell(\hat{\mathbf{y}}_t^i, \mathbf{y}_t^{i*}) = \lambda_\phi (1 - \cos(\hat{\phi}_t^i - \phi_t^{i*})) + \lambda_\rho (\hat{\rho}_t^i - \rho_t^{i*})^2, \quad \mathcal{L}_{\text{aux}} = \mathcal{L}_{\text{geo}} + \mathcal{L}_{\text{vis}}, \quad (4)$$

554 where  $(\hat{\phi}_t^i, \hat{\rho}_t^i)$  are the predicted bearing and range. The visibility BCE is the only training signal  
555 for the visibility head; the actor-side threshold and the detached gate below stop the policy gradient  
556 from reaching it.

557 The actor consumes geometry through a scheduled ground-truth-to-prediction mixture, measured in  
558 policy updates  $k$  (in code, world-size-normalized environment steps with  $k \approx \text{step}/24$ ). Let  $\beta_k$  be  
559 the prediction weight and  $\alpha_k = 1 - \beta_k$  the ground-truth weight. Rather than a linear decay,  $\beta_k$   
560 follows a monotone, concave  $\sqrt{\cdot}$  ramp from a warm-up update  $k_0$  to an end update  $k_1$ ,

$$\beta_k = \max\left(\beta_{k-1}, \sqrt{\text{clip}((k - k_0)/(k_1 - k_0), 0, 1)}\right), \quad \alpha_k = 1 - \beta_k, \quad (5)$$

561 i.e. pure ground truth for  $k \leq k_0$ , ramping to pure prediction at  $k_1$ . The ramp is ratcheted (non-  
562 decreasing) and is frozen whenever the auxiliary-loss EMA exceeds a threshold before a grace up-  
563 date, so the actor is shifted onto its own predictions only as they become reliable. Geometry and

visibility are mixed under this *single* schedule, with the mixed visibility thresholded into a hard indicator:

$$\tilde{\mathbf{y}}_t^i = \alpha_k \mathbf{y}_t^{i*} + (1 - \alpha_k) \hat{\mathbf{y}}_t^i, \quad \tilde{p}_t^i = \alpha_k v_t^{i*} + (1 - \alpha_k) p_t^i, \quad \tilde{m}_t^i = \mathbb{I}[\tilde{p}_t^i > 0.5]. \quad (6)$$

The auxiliary actor feature gates each geometry by a *soft, detached* visibility factor and appends the hard indicator,

$$\text{vis}_t^i = \text{clip}(\text{detach}(\tilde{p}_t^i), \geq \epsilon), \quad \mathbf{a}_t^{\text{aux}} = [\text{vis}_t^i \tilde{\mathbf{y}}_t^i, \tilde{m}_t^i]_i, \quad (7)$$

with a small floor  $\epsilon$  (here 0.05) so the gate never fully zeroes the feature. Detaching the gate keeps it from pulling the visibility head off its BCE target, so the soft gate  $\text{vis}_t^i$  and the appended hard bit  $\tilde{m}_t^i$  play distinct roles. At deployment the mixture is disabled ( $\alpha_k = 0$ ), and all auxiliary actor features are predicted from the depth branch. Figure 13 shows the training curve of GT-annealing.

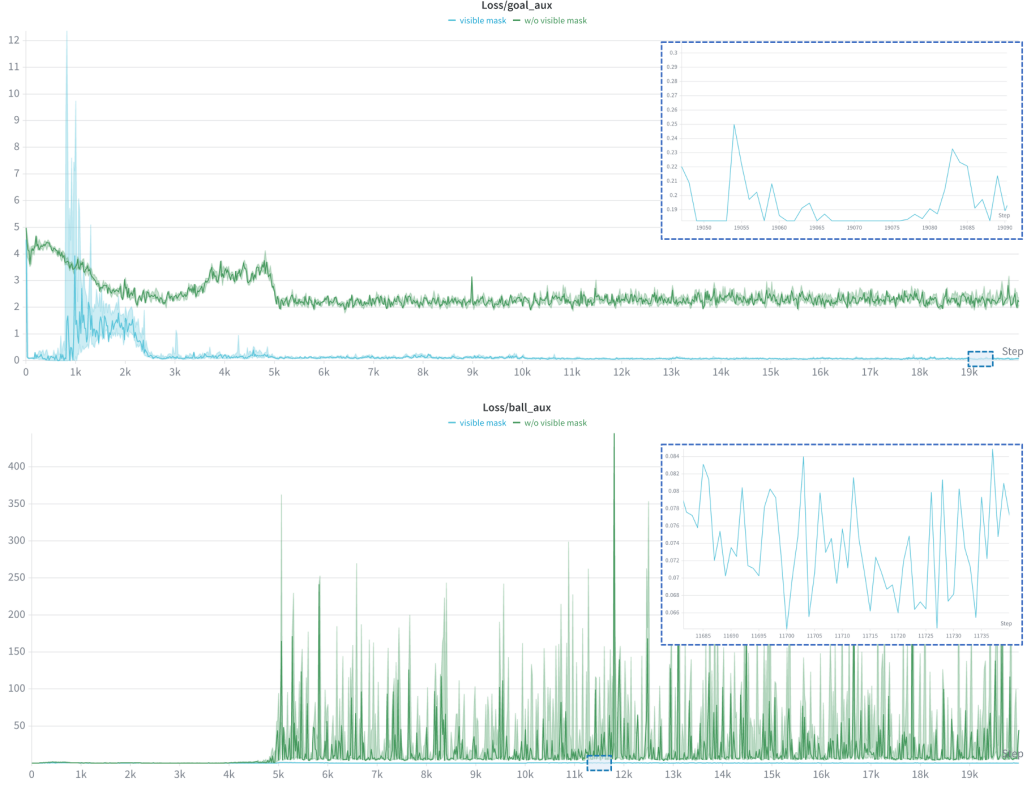


Figure 13: Loss curve of GT-annealing and w/o GT-annealing

## D Task Definitions and Reward Functions

### D.1 Goal-Directed Shooting

Shooting is trained as a goal-directed contact task. Task-specific training uses 1024 parallel environments, a 20 s episode length, a 0.005 s physics step, and a control decimation of the shared deployment and training hardware settings are described in Appendix A.

Let  $d_t$  denote the robot–ball distance, and define the ball-to-goal direction as

$$\mathbf{u}_t^{bg} = \frac{\mathbf{g}_t - \mathbf{b}_t}{\|\mathbf{g}_t - \mathbf{b}_t\|_2}.$$

Near-ball alignment and contact terms are modulated by a smooth distance gate,

$$G(d_t; d_0, \tau) = \sigma\left(\frac{d_0 - d_t}{\tau}\right),$$

which gives larger weights to these terms as the robot approaches the ball. The shooting reward is organized into approach, active vision, ball-goal alignment, contact, terminal success, and safety regularization terms, as summarized in Table 11.

Table 11: Shooting reward summary.

| Reward Terms                    | Components                                                                                                                             | Weight                   |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Approach and chase              | Approach ball reward                                                                                                                   | 3.0                      |
|                                 | Adaptive velocity reward                                                                                                               | 2.0                      |
| Active vision                   | Head track ball reward                                                                                                                 | 3.0                      |
|                                 | Keep in horizon reward                                                                                                                 | -0.5                     |
|                                 | Body follow head reward                                                                                                                | 2.0                      |
| Ball-goal alignment             | Robot behind ball reward                                                                                                               | 1.0                      |
|                                 | Lateral kick line penalty                                                                                                              | 1.0                      |
|                                 | Head yaw align reward                                                                                                                  | 1.5                      |
| Ball reward to goal             | Ankle toward ball reward                                                                                                               | 2.0                      |
|                                 | Ball velocity reward                                                                                                                   | 4.0                      |
|                                 | Ball velocity direction reward                                                                                                         | 1.5                      |
| Sparse goal scored              | One-shot score reward                                                                                                                  | 10.0                     |
| Safety and style regularization | Unstable posture; Foot sliding; Excessive contact force; Joint limits; Joint acceleration; Action-rate changes; Upper body deviations; | task-dependent penalties |
| AMP motion prior                | AMP style reward.                                                                                                                      | coefficient 0.8          |

## D.2 Directional Dribbling

Directional dribbling is trained as sustained ball regulation under a discrete direction command. The robot must keep the ball in a controllable front band while moving it along the commanded direction. We use a 12-way direction command ring,  $\{k\pi/6 : k = -6, \dots, 5\}$ , and update the command with a wrap-around Markov chain over stay,  $\pm 1$ , and  $\pm 2$  bin changes with probabilities 0.40, 0.20, and 0.10, respectively. Each command is held for 4–10 s before resampling. The dribbling episode length is 60 s, with a 0.005 s physics step and a control decimation of the shared deployment and training hardware settings are described in Appendix A.

The controllable band determines whether the ball is under control. A step is in-band when the robot-ball distance is near the 0.42 m target, the ball speed lies in 0.20–1.50 m/s, and the ball velocity does not oppose the commanded direction. During training, the band half-width is tightened from 0.15 m to 0.10 m. The round terminates as loss of ball control if the ball exceeds the 0.80 m far-field scope or remains out of band beyond the curriculum-tightened dwell time of 3.5–6.0 s.

The dribbling reward combines commanded progress, re-approach, visual tracking, heading alignment, out-of-band penalties, safety regularization, and AMP-style motion priors. Table 12 summarizes the reward and regularization groups used for training.

## E Domain Randomization and Sim-to-Real Training

Real-robot transfer is handled by keeping the deployed actor interface fixed and matching simulated observations to this interface during training. The deployed inputs are described in Appendix A, and the policy architecture and optimization details are summarized in Appendix C. This section focuses on the remaining sim-to-real gaps: depth sensing and robot-ball-ground physical variation.

The simulated camera is mounted on the active head and renders `distance_to_image_plane`. Both simulation and real deployment use a single  $168 \times 80$  aligned depth frame, clipped to 0.3–5.0 m, normalized as  $2D/5.0 - 1$ , and zero-filled at invalid pixels. In simulation, this tensor is obtained by rendering a  $168 \times 94$  pinhole depth image with calibrated ZED2i intrinsics and cropping it to the

Table 12: Dribbling reward and regularization summary.

| Group                           | Terms                                                                                                                                  | Weight / setting                   |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Commanded progress & heading    | Ball command reward                                                                                                                    | 5.0                                |
|                                 | Body command reward                                                                                                                    | 0.4                                |
| Ball control                    | Dense re-approach reward                                                                                                               | 2.0                                |
|                                 | Out-of-band penalty                                                                                                                    | -1.0                               |
| Active vision                   | Ball in view reward                                                                                                                    | 1.0                                |
|                                 | Body follow head reward                                                                                                                | 0.6                                |
| Safety and style regularization | Unstable posture; Foot sliding; Excessive contact force; Joint limits; Joint acceleration; Action-rate changes; Upper body deviations; | task-dependent penalties           |
| Termination                     | Falls penalty; Bad-orientation penalty; Lose ball control penalty;                                                                     | -1.0                               |
| AMP motion prior                | Head-masked Wasserstein AMP style reward                                                                                               | coefficient 0.8, interpolation 0.8 |

actor input size. The real controller applies the same cropping, range filtering, and normalization before actor inference. Temporal information is provided by proprioceptive history rather than by stacking depth frames.

During training, we randomize visual observations and physical interaction parameters. Depth perturbations are applied to the rendered tensor, while robot, ball, ground, and contact variations are applied through IsaacLab event terms at startup, reset, or fixed intervals. Table 13 summarizes the perturbation groups used for sim-to-real transfer.

Table 13: Representative training randomization ranges for sim-to-real transfer.

| Perturbation                           | Range / value                                   |
|----------------------------------------|-------------------------------------------------|
| Valid depth range                      | 0.3–5.0 m                                       |
| Distance-proportional depth noise      | 0.01                                            |
| Gaussian depth-noise coefficient       | 0.005                                           |
| Salt-and-pepper probability            | 0.01                                            |
| Stripe dropout probability             | 0.005                                           |
| Edge-drag probability                  | 0.01                                            |
| Rectangular mask probability           | 0.1                                             |
| Rectangular mask size                  | 10–25%                                          |
| Ball-region occlusion probability      | 0.2                                             |
| Ball-region occlusion radius           | 2–10 px                                         |
| Partial occlusion ratio                | 45–85%                                          |
| Border near-field noise probability    | 0.15                                            |
| Camera intrinsic perturbation          | $\pm 0.3$ – $\pm 0.5$ px                        |
| Camera translation perturbation        | 0.02–0.04 m                                     |
| Camera rotation perturbation           | 0.025–0.060 rad                                 |
| Joint-position offset                  | $\pm 0.02$ rad                                  |
| Base COM perturbation                  | $\pm 0.05$ m                                    |
| Base mass perturbation                 | $\pm 5$ kg                                      |
| Actuator stiffness / damping scale     | 0.8–1.2                                         |
| Ball mass randomization                | 0.36–0.56 kg                                    |
| External push interval                 | 3–15 s                                          |
| External linear velocity perturbation  | $x, y \in [-1, 1]$ m/s; $z \in [-0.2, 0.2]$ m/s |
| External angular velocity perturbation | $[-0.78, 0.78]$ rad/s                           |

## F Additional Results

This section contains additional empirical results that complement the main paper.

### F.1 Powerful Shot.

We additionally study a close-range powerful-shot skill in simulation. This skill uses the same depth-only actor interface and auxiliary ball-geometry prediction as the base skills; only the reset distribution, reward terms, and motion-prior schedule are changed.

We use the same near-ball reward gating as in the shooting task. The reward follows a two-stage curriculum: the first stage rewards near-ball leg effort through a distance-weighted foot-speed term, and the second stage adds ball-outcome terms, including horizontal ball speed and displacement from the reset position. The corresponding reward scales are ramped in sequentially during training.

An AMP kick-motion prior is activated once a sufficient fraction of parallel environments reaches the near-ball zone, regularizing the explosive leg swing while keeping the actor input unchanged. This experiment tests whether the same depth-only framework can support high-speed striking from a close pre-kick stance without adding a separate runtime ball-localization or shooting controller. Figure 14 showed the real world result.



Figure 14: Power shoot real-world example

### F.2 Dribbling while Avoiding Obstacles.

This skill highlights the extensibility of our depth-predicted auxiliary heads: extending directional dribbling to avoid obstacles requires no architectural change. Following the scalable visibility-aware auxiliary geometry of Sec. 3.3, the obstacle simply enters the task object set as one more object, adding a single indexed geometry/visibility head that predicts its relative position and visibility from the shared depth latent—with no privileged obstacle input and no change to the actor interface or control loop.

Avoidance is induced through the reward rather than a separate controller: an artificial potential field deflects the commanded direction  $\mathbf{d}_t^{\text{cmd}}$  by a tangential repulsion from the nearest in-path obstacle,

$$\tilde{\mathbf{d}}_t^{\text{cmd}} = \frac{\mathbf{d}_t^{\text{cmd}} + \kappa w_t a_t \mathbf{t}_t}{\|\mathbf{d}_t^{\text{cmd}} + \kappa w_t a_t \mathbf{t}_t\|}, \quad (8)$$

where  $\mathbf{t}_t$  is the unit normal to the command on the obstacle-clearing side ( $\mathbf{t}_t \cdot \mathbf{u}_t \leq 0$ , with  $\mathbf{u}_t$  the ball-to-obstacle bearing),  $a_t = \max(\mathbf{u}_t \cdot \mathbf{d}_t^{\text{cmd}}, 0)$  activates the repulsion only for obstacles ahead,  $w_t \in [0, 1]$  is a proximity weight that vanishes beyond an influence radius, and  $\kappa$  is the steering gain. Every direction-aware dribble reward tracks  $\tilde{\mathbf{d}}_t^{\text{cmd}}$  in place of  $\mathbf{d}_t^{\text{cmd}}$ , so obstacle avoidance emerges from the same velocity-direction tracking objective with no change to the reward structure. We show detailed loss curve of the aux loss in Figure 15.

### F.3 Recovery upon Losing the Ball.

We also test whether the chase policy can recover after the ball temporarily leaves the head-mounted field of view. The recovery behavior is not implemented as a separate search controller. Instead, the

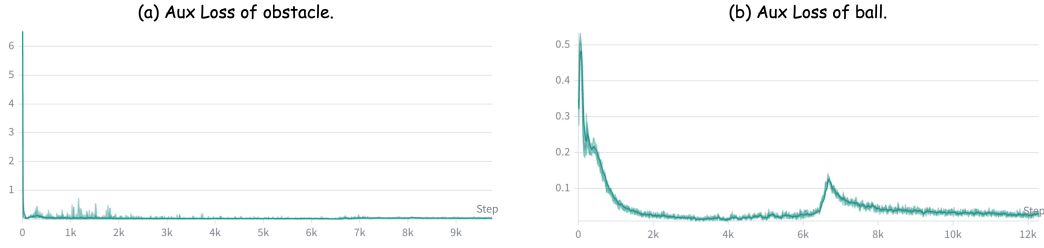


Figure 15: Ball and obstacles’ Aux loss: The prediction heads for obstacles and soccer can both converge very well.

647 same active-vision policy is shaped by a visibility flag and an invisible-ball timer: when the ball is  
 648 visible, the robot continues pursuit; when it is lost, the policy is penalized for prolonged invisibility  
 649 and for moving too fast without reliable visual evidence. In the submitted configuration, the search  
 650 penalty starts after 0.5 s of continuous invisibility with weight 0.2, and the invisible-speed penalty  
 651 limits body-frame forward speed around 0.5 m/s with weight 0.6.

652 Recovery is driven by the active head and body-following rewards. The head reward tracks the de-  
 653 sired yaw/pitch angles toward the ball in the robot body frame, while the body-follow-head reward  
 654 encourages the torso to realign with the recovered viewing direction. These rewards are introduced  
 655 gradually: the head reward starts at 12k steps and ramps over 36k steps, and the body-following re-  
 656 ward starts at 48k steps and ramps over 48k steps. This produces a closed-loop search-and-reacquire  
 657 behavior while preserving the same depth-only actor interface used by the main soccer skills.